On Market Making for a New Exchange: EUR/USD Designated Market Maker on Exchange A

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1 Introduction

1.1 Motivation

A newly launched exchange, referred as exchange A, has mandated our firm as the sole Designated Market Maker for its EUR/USD spot book. Two established venues, B and C, remain the primary sites of price discovery, carrying on average 75% and 25% of the daily volume respectively. We receive market data from B with a 200 ms delay and from C with a 170 ms delay, while high-frequency participants reach the same data in 50 ms and can act on exchange A before our own view of fair value has refreshed. The mandate commits us to continuously quoting ten levels on each side of the book at a common 1 bp tick, with a starting capital K split evenly between EUR and USD and an asymmetric fee grid. A delta limit caps the EUR inventory we are willing to warehouse before externalising risk as taker flow on B and C.

Three constraints shape every design decision that follows. First, on the day exchange A goes live no historical fills or counterparty behaviour are observable, so the initial quoting policy cannot be learnt from data and must be derived from first-principles microstructure models. Second, the latency gap to HFTs exposes every resting quote to adverse selection, which the pricing model must compensate for rather than ignore. Third, because we cannot market-make on B or C, the delta limit turns inventory management into an active cross-venue hedging problem rather than a passive book-balancing exercise.

1.2 Objective

Our goal is to deliver an end-to-end market-making system that addresses the three constraints above and can be evaluated under the conditions described in the subject brief. The project is split into three phases matching the mandate timeline. Phase 1 bootstraps liquidity without any trading history, using a heuristic quoting policy derived from classical market-making theory. The fair mid is reconstructed from the two reference venues at their respective latencies, the bid-ask spread is sized to compensate for both inventory risk and adverse selection, and quotes are skewed in the direction of recent order-flow imbalance. A dedicated hedge router externalises excess EUR inventory on B and C whenever the delta limit is approached. Phase 2 re-estimates the Phase 1 parameters by maximum likelihood from a simulated month of fills. Phase 3 studies how our quoting policy should degrade from a price-competitive quoter to a reliable fallback liquidity provider once aggressive HFT market makers populate the inside of the book.

1.3 Simulation Environment

Because the exchange has no history at launch and we cannot run the strategy against real counterparties, the evaluation environment is a layered simulation built specifically for this project. A stochastic mid-price generator (GBM, GARCH with jumps, or Heston) produces the fair value underlying both reference venues. A venue simulator wraps that path into a realistic bid/ask and depth profile per venue, with a choice of five spread regimes (static, stochastic, adaptive, asymmetric, and skewed) reflecting the microstructure assumptions we want to stress. A limit order book with a matching engine and fill events sits at the centre of exchange A; a Poisson client-flow generator with truncated Pareto sizes delivers counterparty orders whose intensity decays exponentially in

the quoted half-spread; and a reporting layer reconstructs the full mark-to-market and realised P&L decomposition from the trade history.

1.4 Evaluation

Each phase is evaluated on the subject-mandated one-month simulated backtest, with the capital K reset to its initial split at the start of every phase. For Phase 1 we report realised and mark-to-market P&L, the percentile envelope of mark-to-market as a function of time since trade inception, top-ten trades by size, full-versus-partial fill rates at each of the ten quoting levels, and inventory trajectories against the delta limit. For Phase 2 we compare the heuristic and calibrated configurations on an independent month using the same diagnostics, so that any difference in performance is attributable to the updated parameters rather than to the market regime. For Phase 3 we measure both our own P&L under HFT competition and the quality of liquidity observed by exchange-A clients, distinguishing small from large trades and quantifying the continuity of two-sided quoting through volatility spikes, one-sided HFT retreats, and venue outages.

1.5 Report Structure

The remainder of the report follows the step-by-step structure of the subject. section 3 introduces the order-book data structure and matching rule; section 4 describes the venue simulator and the family of spread regimes supported; section 5 presents the quoting framework in full; section 7 details the inventory and P&L bookkeeping; section 8 gathers the Phase 1 backtest outputs; section 9 analyses the calibration results; section 10 examines the Phase 3 fallback behaviour; and section 11 summarises the findings and outlines the remaining modelling questions.

2 Literature Review

The design of an inventory-aware quoting policy originates with Ho and Stoll [11], who frame the dealer's problem as a trade-off between expected spread capture and the variance of an accumulating inventory. Avellaneda and Stoikov [2] recast this intuition as a stochastic control problem with a Poisson fill intensity of the form $\lambda(\delta) = A\,e^{-k\delta}$ and derive closed-form reservation prices that tilt quotes against the current position. Because the original Avellaneda–Stoikov solution depends on a terminal liquidation date that does not exist for a 24/5 FX book, we follow the infinite-horizon limit described by Guéant et al. [10] and Guéant [9], in which a discount rate ω replaces the time-to-horizon and the inventory penalty becomes stationary. This is the formulation implemented in our Phase 1 quoter.

The second strand of the literature informs our treatment of order flow and adverse selection. Roll [13] gives the effective-spread decomposition we rely on for cross-venue sanity checks, while Biais et al. [3] provide the empirical baseline for limit order book dynamics. Engle and Patton [7] and Cont et al. [6] establish that short-horizon quote revisions are driven primarily by signed order-flow imbalance; this motivates our OFI-based skew, which adjusts the reservation price in the direction of recent imbalance rather than inferring it from price moves alone. To compensate quoted spreads for the 150 ms latency gap against HFTs, we draw on the adverse-selection and latency-aware quoting

refinements consolidated by Cartea et al. [5], which justify the asymmetric spread regime and the emergency inventory multiplier used in our hedge controller.

The Phase 2 calibration layer re-estimates the statistical primitives that Phase 1 took as given. Volatility enters the quoter through the A–S inventory term $\gamma\sigma^2/(2\omega)$; we estimate σ with a GARCH(1,1) model [4] on the simulated reference mid, and cross-check it against the exponentially weighted estimator of J.P. Morgan/Reuters [12], which is better suited to the short effective horizons over which the quoter reacts. Client order sizes are heavy-tailed but bounded from above, so we fit a truncated Pareto distribution following Aban et al. [1], whose MLE formulation handles the upper cap cleanly; the power-law regularity of large trades documented by Gabaix et al. [8] is what justifies this parametric family in the first place.

Relative to this body of work, our contribution is a fully simulated multi-venue FX designated-market-maker study rather than a new theoretical model. We combine an infinite-horizon Avellaneda–Stoikov skeleton with an OFI-based skew, a cross-venue latency-aware fair mid that blends two reference venues at their respective delays, session-adaptive arrival decay, and a depth-gated hedge router that externalises inventory on the reference venues. The whole pipeline is run in two stages: a heuristic-parameter Phase 1, followed by a Phase 2 in which arrival intensity, size distribution, volatility, and the risk-aversion γ are re-estimated by maximum likelihood on a simulated month of fills. This two-phase calibration workflow lets us quantify the marginal value of data-driven parameters over first-principles defaults in an environment that would otherwise have no trading history at launch.

3 Order Book

The order book of exchange A must show a price and a quantity at ten levels on both sides at all times. This section describes the data structure we chose to hold that state, how submissions and cancellations flow through it, and the invariants of the matching engine. The design is exercised on a realistic EUR/USD seed (mid = 1.10, 1 bp tick, ten levels per side, 1M EUR per level).

3.1 Data structure

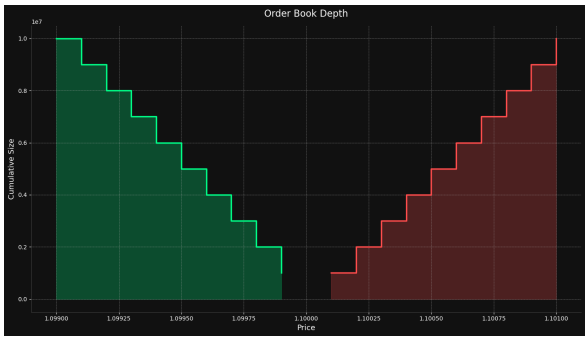
Two pressures shape the design. Matching requires a price-time-ordered view of the market-maker (MM) side at clearing time, yet the hot path is dominated by bulk requotes — up to ten cancels followed by ten posts per step. Keeping the book permanently sorted would pay on every mutation for a benefit consumed only once per step.

We therefore split orders by origin (MM versus client) into two hash-indexed stores and keep a price-ordered view of the MM side as a lazy cache, rebuilt only when the matcher runs. Cancels and posts touch only the hash stores; the MM side is sorted once per clearing call instead of being re-ordered on every mutation. Time priority within a price level is preserved by a counter that grows by one on every insertion, so later orders carry higher numbers than earlier ones.

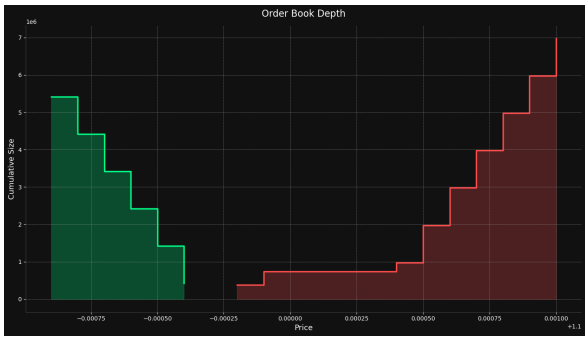
3.2 Submissions and cancellations

The two sides are handled asymmetrically. MM cancels and posts arrive in bulk and never run the matcher, because passive quotes cannot cross by construction. Client orders are

inserted one at a time, and the matcher runs immediately after each insertion.



(a) Seeded book: one order per level, symmetric depth.



(b) After twenty marketable client orders: shallow levels consumed from the top.

Figure 1: Cumulative depth before and after a burst of client flow.

3.3 Matching rule

Three invariants govern every clearing call. Client orders only match against MM orders, never against each other: as the sole designated market maker for Phases 1 and 2 we are on one side of every trade. The fill price is always the passive MM price, never the incoming aggressive price — clients pay the spread we quote, which is the only source of gross revenue. And matching follows strict price-time priority with partial fills: the engine walks the sorted MM side, deducts the minimum of the two sizes, logs the match, and either retires the filled order or leaves the remnant resting. The residual shape in Figure 1 is a direct visual check of this rule: the top of the book is consumed first, deeper levels are untouched.

3.4 Fills as events

Every match is dispatched synchronously, as a fill event, to a single registered listener. In the full simulation the listener is the Quoter, which updates inventory, trade history, and the pending top-up queue inside the clearing call. The Quoter therefore never polls the book: the next quote computation already sees the post-fill inventory, which removes staleness bugs on steps that both fill and requote. Decoupling the book from its consumer also keeps the same engine usable in Phase 3, where a separate diagnostic listener records fills for the exchange-client-side liquidity analysis without touching the Phase 1 code path.

4 Market Simulator for Venues B and C

Exchanges B and C are external reference venues: we never post on them, we only read them. Our quoting engine on A consumes the best bid, the best ask, and the best-size of each, and nothing else. We therefore do not simulate a full limit order book on B or C: all the depth beyond the top of book would be detail the rest of the system never looks at. Instead we simulate a mid-price process and build a synthetic bid/ask/depth layer around it. This is the second of the two options offered by the subject, and it is both cheaper to run and sufficient for everything the market maker on A has to decide. The

reduced-form choice keeps realism where it matters — spreads that widen under stress and depth that thins with them — while avoiding the combinatorial cost of simulating individual order arrivals at two venues.

4.1 Noised mid

The fundamental value V_t is the GBM (or GARCH, or Heston) path produced by the stock simulator. The observed mid on B and C is V_t plus microstructure noise, in the spirit of Roll [13]:

$$P^{\text{mid}}_{B/C}(t) = V_t + \eta \cdot \sigma \cdot \sqrt{\Delta t} \cdot Z_t, \quad Z_t \sim \mathcal{N}(0,1),$$

with $\eta = 0.1$. Each venue draws an independent Z_t path, so B and C are noised versions of the same fundamental. One subtle point: every spread model below estimates realized volatility from V_t, not from P^{mid}, because differencing the noised series would double-count the microstructure variance inside the vol estimator. The zero-mean estimator used for that (see `compute_rv_zero_mean`) is unbiased for zero-drift GBM and lower-variance than the sample std over short windows.

4.2 Spread regimes

We benchmarked five regimes of increasing realism — Static, Stochastic, Adaptive, Asymmetric, and Skewed — and kept two. We reject Static because a price-independent width ignores volatility entirely; the market maker is widest exactly when they should be widest, but never adapts. Stochastic scales the width with realized vol but never returns to the competitive floor even in calm markets, which systematically over-charges. Asymmetric mean-reverts realistically but treats up-moves and down-moves the same way, leaving a latency-arbitrage gap for HFTs who, seeing the vol spike 150 ms before us, can infer our quote path. We deploy **Adaptive on** C and **Skewed on** B.

Venue C **— Adaptive.** A symmetric half-spread proportional to how much realized vol deviates from the long-run input σ:

$$h(t) \;=\; h_0 \cdot \Big(1 + \alpha \cdot \big(\widehat{RV}_t/\sigma - 1\big)\Big), \qquad h(t) \geq 0.1 \cdot h_0,$$

with $h_0 = \text{mid}(t) \cdot s_{\text{bps}}/10000$ and $\widehat{RV}_t$ an EMA-smoothed rolling vol over a 120 s window. The width is tight at baseline, widens on shocks, and tightens symmetrically when calm returns. A 10% floor prevents collapse near zero. C carries 25% of volume — a smaller venue where informed flow is weaker, competition enforces a tight floor, and the symmetry of up vs. down adjustment is not a material leak.

Venue B **— Skewed.** Same vol-adaptive width, plus a directional shift of the quote centre driven by a short-horizon momentum signal. Let $\hat{m}_t$ be the EMA of log-returns normalised by its steady-state noise floor; the skew is

$$\delta_t \;=\; \gamma \cdot h(t) \cdot \tanh\big(\text{sign}(\hat{m}_t) \cdot \max(|\hat{m}_t| - \tau,\, 0)\big),$$

with $\tau = 1.5$ the dead-band threshold. Small random wiggles map to $\delta_t = 0$ exactly; only sustained directional moves shift the centre. Posted quotes become ask $=$ mid $+ h + \delta$ and

bid $=$ mid $- h + \delta$: width is unchanged, the centre moves in the direction of the trend. B carries the dominant 75% of volume, where short-horizon order-flow imbalance predicts returns [6]; a momentum-unaware venue there would be repeatedly picked off [3, 7].

4.3 Depth

Depth at each step is derived deterministically from spread width, not simulated as an independent process:

$$\text{depth}_t \;=\; \bar{D} \cdot \frac{\overline{S}}{S_t},$$

where S_t is the full spread and $\overline{S}$ its session mean. The economic story is simple: when the external market maker is confident, they quote tight and large; when stressed, they pull size and widen. The hedge router on A reads this array with the same 200 ms (B) / 170 ms (C) latency offset as the mid, and caps each hedge leg at the available depth.

4.4 One stock, two venues

B and C are two `Market` objects wrapping the same `Stock` via `copy.deepcopy`. The fundamental path V_t is shared; the noise draws and spread realisations are independent. We deliberately do not model B/C price divergence: the cross-venue edge in this project is the latency asymmetry (HFTs see B/C at 50 ms while we see them at 200/170 ms), not price disagreement between venues.

4.5 Validation

Figure 2 overlays the four retained regimes on a shared GARCH-plus-jump run — the one fundamental where the models visibly differentiate. The top two panels plot ask and bid quotes for Static, Stochastic, Adaptive, and Skewed against the common mid; the four middle panels isolate each width path with a Pearson correlation to realized vol annotated in the title; the bottom panel shows the skew δ_t activating around the jump under the Skewed regime and remaining near zero elsewhere. The Static width is flat up to the mid-scaling factor, while Stochastic, Adaptive, and Skewed all co-move with $\widehat{RV}_t$ ($\rho \approx 1.00$, 1.00, 0.97), confirming the vol-adaptive mechanism fires as intended.

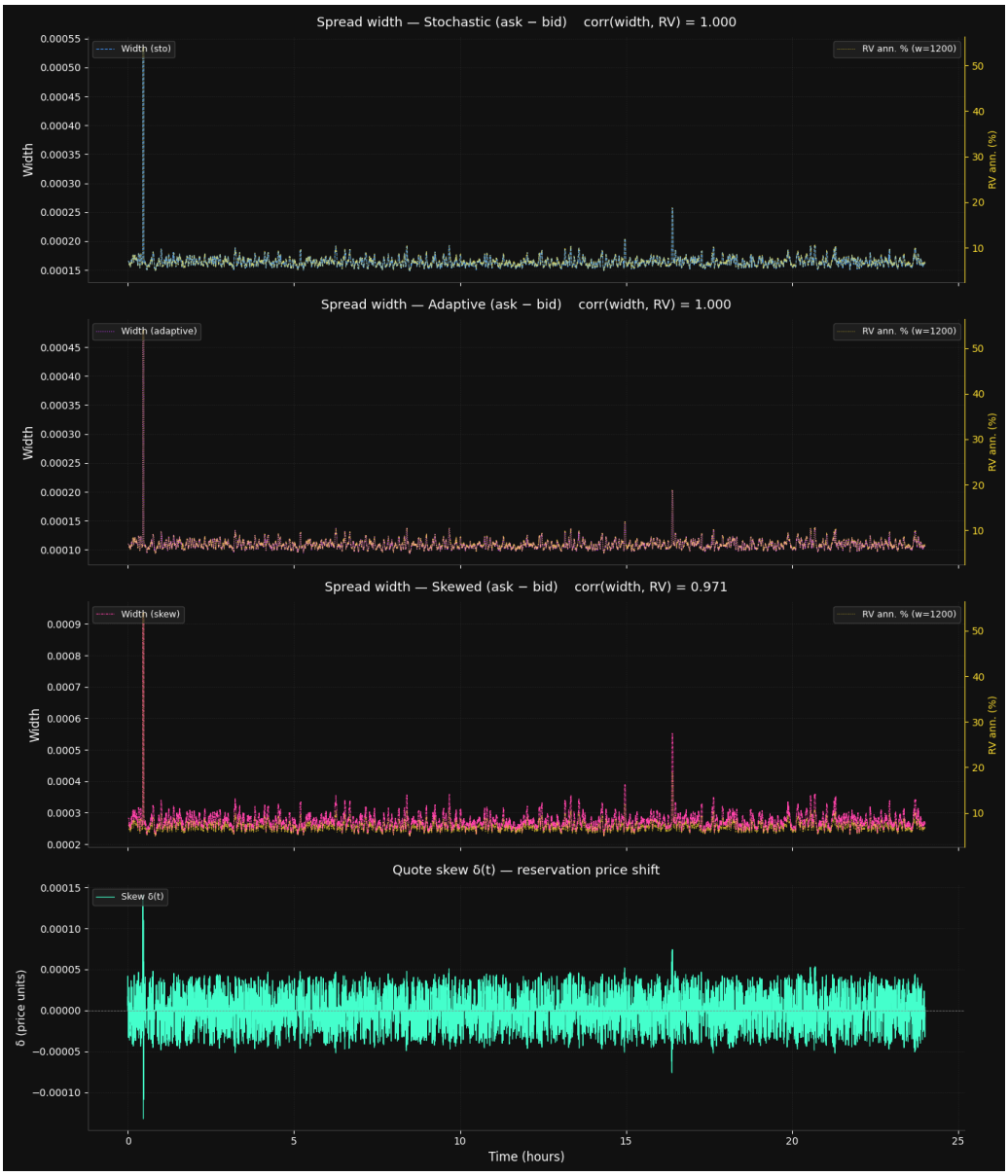


Figure 2: Ask and bid quotes, width panels with realized-vol overlay, and the Skew δ_t path on a shared GARCH+jump run.

5 Market-Making Logic on Exchange A

Exchange A has no history, so every quote we post is derived from the state of B, C, our inventory q, and our capital K. Given these inputs the quoter must output two things each step: the set of orders to post and the set of resting orders to cancel. We describe here the modelling decisions that produce those two lists, not the Python that assembles them. The six subsections follow the order in which the decisions are composed: read the reference price, compute a reservation price, widen a spread around it, skew asymmetrically and lay down a ladder, decide whether to re-quote at all, and finally hedge when inventory breaches its limit.

5.1 Fair mid under latency asymmetry

We never quote against A's own mid: A has no informative price until we seed one. The reference is instead a cross-venue best constructed from B and C read with their respective latency offsets $\ell_B = 200\,\text{ms}$ and $\ell_C = 170\,\text{ms}$,

$$\text{bid}_t^{\text{fair}} = \max\big(\text{bid}_B(t-\ell_B),\ \text{bid}_C(t-\ell_C)\big), \qquad \text{ask}_t^{\text{fair}} = \min\big(\text{ask}_B(t-\ell_B),\ \text{ask}_C(t-\ell_C)\big),$$

with $\text{mid}_t^{\text{fair}} = \frac{1}{2}(\text{bid}_t^{\text{fair}} + \text{ask}_t^{\text{fair}})$. The choice of the cross-venue *best* rather than a volume-weighted average is a defence against being arbitraged: clients can always transact at the tighter side elsewhere, so anything wider on A is free optionality handed to adversarial flow. HFTs read the same feeds at $\ell_{\text{HFT}} = 50\,\text{ms}$, so the price we quote against is always structurally stale relative to theirs; this gap is priced explicitly as a spread component in Section 5.3.

5.2 Reservation price: Avellaneda–Stoikov with infinite horizon

The classical inventory-shift mechanism of Avellaneda and Stoikov [2] uses a finite horizon T with the penalty $\gamma\sigma^2(T-t)$. That formulation is inappropriate for a 24/5 FX market: there is no terminal liquidation time at which inventory becomes worthless. We replace $T-t$ by a stationary discount $1/\omega$ with $\omega = 1/(8\,\text{h})$ — one FX session as the effective risk horizon — which is the infinite-horizon limit of the same HJB system, studied formally by Guéant et al. [10].

A second correction is numerical. Applying the raw Avellaneda–Stoikov shift with inventory in EUR (range 0–900k) produces a reservation displacement of several hundred basis points at ten percent inventory, three orders of magnitude larger than the half-spread itself. We therefore work with the normalised ratio $\rho = q/K$ and express the shift as a fraction of the half-spread, so that at $|\rho| = 1$ the quoted centre is displaced by exactly one half-spread:

$$r_t \;=\; \text{mid}_t^{\text{fair}} \;-\; \rho \cdot h_t.$$

Direction is preserved (a long EUR position drags the centre down and attracts sells); magnitude is now calibrated to the spread itself, with no new free parameter beyond γ.

5.3 Three-component spread with fee floor

The half-spread h_t is built as the sum of three economically distinct adverse-selection premia, floored by a fee-cover term,

$$h_t \;=\; \tfrac{1}{2}\big(h_t^{\text{AS}} + h_t^{\text{lat}} + h_t^{\text{inv}}\big), \qquad h_t \;\geq\; \phi \cdot \text{fee}_{\text{maker}}^{A} \cdot \text{mid}_t^{\text{fair}}.$$

The Avellaneda–Stoikov component

$$h_t^{\text{AS}}(\text{bps}) \;=\; \gamma \cdot (\sigma \cdot 100)^2 \cdot \tfrac{1}{\omega} \;+\; \tfrac{2}{\gamma}\,\ln\big(1 + \gamma/k\big)$$

combines inventory-risk aversion (first term) with the convexity of the exponential order-arrival intensity (second term), as derived by Avellaneda and Stoikov [2] after the infinite-horizon substitution. The latency term $h_t^{\text{lat}} = 2\sigma_{\text{sec}}\sqrt{\Delta\ell}$, with $\Delta\ell = w_B(\ell_B - \ell_{\text{HFT}}) + w_C(\ell_C - \ell_{\text{HFT}})$, is two standard deviations of the expected adverse move during the window in which HFTs see the reference feeds and we do not — a cushion against the free option they hold against our stale quotes. The capacity term $h_t^{\text{inv}} = \alpha_s \rho^2 h_t^{\text{AS}}$ is quadratic in the

inventory ratio: it is negligible near flat and widens sharply as we approach the delta limit, pushing flow away from the side that would deepen our position. Finally the floor $\phi \cdot \mathrm{fee}^{A}_{\mathrm{maker}} \cdot \mathrm{mid}^{\mathrm{fair}}_{t}$ guarantees that the inception spread covers the maker fee on every fill: no trade is ever gross-negative, only fee-negative offset by subsequent inventory P&L. The floor binds in calm regimes where the three adversarial premia would otherwise collapse below the fee.

5.4 Asymmetric skew and ladder

The reservation shift in Section 5.2 is symmetric: it moves both sides of the quote by the same amount. Guéant [9] shows that the asymptotic optimum is instead an *asymmetric* split of the half-spread — widen the side that would deepen inventory, tighten the side that would reduce it. We therefore apply the Guéant skew

$$\Delta_t \;=\; q \cdot \sqrt{\frac{\gamma\sigma^2}{2k}}, \qquad \mathrm{ask}^{\mathrm{post}}_t = r_t + h_t - \Delta_t, \qquad \mathrm{bid}^{\mathrm{post}}_t = r_t - h_t - \Delta_t,$$

with Δ_t clipped so the tighter side can never breach the fee floor.

The subject requires ten levels on each side. Levels step one tick further from the mid at each rung; sizes decay exponentially and are skewed against inventory,

$$\mathrm{size}^{\mathrm{bid}}_i \;=\; Q_0\, e^{-\beta i}\,(1 - \tfrac{1}{2}\overline{\rho}), \qquad \mathrm{size}^{\mathrm{ask}}_i \;=\; Q_0\, e^{-\beta i}\,(1 + \tfrac{1}{2}\overline{\rho}),$$

with $\overline{\rho} = \mathrm{clip}(\rho, -1, 1)$. This is a second, complementary inventory correction acting through *depth* rather than *price*: even if the Guéant skew is clipped by the fee floor, the ladder sizes still bias against further accumulation. Exponential decay concentrates displayed liquidity at the top of book, consistent with the empirical shape of limit-order-book queues.

5.5 Selective re-quoting

Re-quoting every step is expensive in cancel/post messages and costly in time priority: orders that are still valid lose their queue position on every spurious cancel. We use a strict six-priority hierarchy, with each lower priority firing only if none above it did (Table 1). Priority 2 treats the 08:00 and 16:00 UTC FX session boundaries as hard resets of the full ladder, so session-specific state does not accumulate across regimes. Priority 3 closes the price-drift leak; priority 5 closes a subtler silent-accumulation leak where market prices are flat but our own inventory has moved enough to invalidate the ladder skew. Priority 6 fires on pure staleness and re-quotes at a tighter 0.7× spread, so the refreshed quotes compete back into relevance rather than being re-posted at the same passive distance.

P	Trigger	Action
1	First call	Full ladder, no cancels
2	Session boundary (08:00/16:00 UTC)	Cancel all, full ladder
3	$\lvert$best-price drift$\rvert > 0.25$ spread	Cancel all, full ladder
4	Fills / top-ups pending	Rebuild filled slots only
5	$\lvert\Delta q\rvert/K > 5\%$ since last reprice	Cancel all, full ladder
6	Order older than τ_{stale}	Cancel stale, reprice at 0.7× spread

Table 1: Requote priorities; at most one rule fires per step.

5.6 Hedge router

When either $|q| > \delta_{\text{lim}} \cdot K/2$ or $|q \cdot \text{mid}^{\text{fair}}| > \delta_{\text{lim}} \cdot K/2$ — that is, either the EUR leg or its USD notional breaches the delta limit on half-capital — we externalise risk through taker market orders on B and C. Three decisions shape the router. First, the USD notional of EUR inventory is valued at the B/C cross-venue mid, not at mid_A, because A's mid is polluted by our own skewed quotes. Second, the split between venues is depth-aware: each venue is scored by

$$s_v \;=\; \frac{1}{\text{fee}^v_{\text{taker}} \;+\; \sigma_{\text{sec}} \cdot \ell_v \;+\; \lambda \cdot \text{mid}^{\text{fair}}/D_v},$$

combining the taker fee, the expected adverse move during execution lag, and a market-impact penalty inversely proportional to the available depth D_v at the lagged step. The hedge is split $s_B : (s_B + s_C)$ with overflow re-routed to the other venue when the preferred one is depth-constrained. Third, when post-hedge inventory remains above the tighter threshold hedge_partial_limit$\cdot K/2$ — meaning the combined $B{+}C$ depth was insufficient to bring us back — we flip an emergency flag that multiplies the Avellaneda–Stoikov inventory-risk term by $5\times$ on the next step. Our own quotes on A become so aggressively asymmetric that organic client flow cleans up the residual. This is the Guéant et al. [10] inventory-penalty mechanism used as an overflow valve: we deliberately do not chase depth we cannot reach.

6 Client Order Flow on Exchange A

The quoter of Section 5 sets prices, but the only channel through which those prices become realised revenue is the stream of client orders reaching Exchange A. We therefore need a generative model of that flow with three layers: *when* an order arrives, *which type* it is, and *how large* it is. The intensity baseline A, the price sensitivity k, the tail index α of order sizes and the market-order mix p_{MO} are design parameters of the simulator: they are held fixed across Phase 1 and Phase 2 and are not re-estimated by the calibration pipeline of Section 9. What Phase 2 recalibrates are the quoter's own risk parameters — the client-flow knobs above remain exogenous throughout.

6.1 Poisson arrivals with price-sensitive intensity

Buy-side and sell-side clients are modelled as two independent inhomogeneous Poisson processes whose instantaneous rates depend on how far our posted quote sits from the prevailing mid on the relevant side. Tightening the ask attracts buy-side flow without mechanically also attracting sell-side flow, which matches the empirical asymmetry of electronic FX and preserves the separation of controls the quoter relies on. With the mid m_t and our top-of-book $(\text{bid}_t, \text{ask}_t)$, the per-side distances in basis points are

$$\delta_t^{\text{ask}} = 10^4\,\frac{\text{ask}_t - m_t}{m_t}, \qquad \delta_t^{\text{bid}} = 10^4\,\frac{m_t - \text{bid}_t}{m_t},$$

and the arrival intensities take the exponential kernel of Avellaneda and Stoikov [2],

$$\lambda_t^{\text{buy}} \;=\; A\,\exp\!\big(-k\,\delta_t^{\text{ask}}\big), \qquad \lambda_t^{\text{sell}} \;=\; A\,\exp\!\big(-k\,\delta_t^{\text{bid}}\big).$$

Over a simulation step of length Δt the number of arrivals on each side is drawn independently as $N^{\text{side}} \sim \text{Poisson}(\lambda_t^{\text{side}} \cdot \Delta t)$, so a step can produce zero, one, or several orders per side. With the defaults $A = 7 \cdot 10^{-3}\,\text{s}^{-1}$ and $k = 0.3\,\text{bp}^{-1}$, the half-life of the intensity is $\ln(2)/k \approx 2.3\,\text{bp}$: a quote posted two bps outside the fair mid already attracts roughly half the arrivals that a touch-priced quote would.

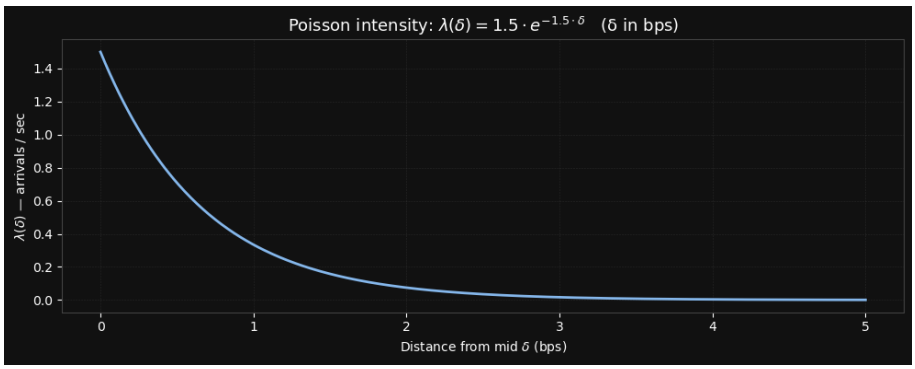


Figure 3: Arrival intensity $\lambda(\delta) = A\,e^{-k\delta}$ as a function of the distance from the fair mid, with $A = 0.007\,\text{s}^{-1}$ and $k = 0.3\,\text{bp}^{-1}$. Each bp of extra distance costs the quote approximately 26% of its arrival rate.

6.2 Market-versus-limit mix and limit placement

Each arrival is independently classified market or limit by a Bernoulli trial with probability $p_{\text{MO}} = 0.5$. Market orders cross immediately at the touch — buys at ask_t, sells at bid_t — and are the mechanism through which the quoter earns its inception spread. Limit orders rest inside the book at a random offset from the mid,

$$p^{\text{LO, buy}} \;=\; m_t\left(1 - \tfrac{\varepsilon}{10^4}\right), \qquad p^{\text{LO, sell}} \;=\; m_t\left(1 + \tfrac{\varepsilon}{10^4}\right), \qquad \varepsilon \sim \text{Exp}(k).$$

Using the same k in both the arrival intensity and the limit-placement distribution is a self-consistency choice rather than a fit: clients who are dissuaded from crossing at rate k are assumed to post passively at the same rate k. This keeps the flow symmetric under a shift of the quoted distance — widening our spread by $\Delta\delta$ reduces arrivals by $e^{-k\Delta\delta}$ and simultaneously pushes the average passive limit deeper by the same factor, with no second free parameter to tune.

6.3 Heavy-tailed order sizes

Client notional sizes are drawn from a truncated Pareto on the interval $[x_{\min}, x_{\max}] = [10^3, 10^5]$ EUR with tail index $\alpha = 1.5$, in line with the power-law volume distribution of Gabaix et al. [8]. Truncation is not only empirical realism — the exchange rejects orders outside its permitted notional band — but also a numerical safeguard: an untruncated Pareto with $\alpha = 1.5$ has infinite variance, and Monte-Carlo P&L moments would be dominated by a handful of outliers instead of by the structural behaviour of the quoter. The truncated inverse-CDF sampler is

$$X \;=\; x_{\min}\left(1 - U\big(1 - (x_{\min}/x_{\max})^{\alpha}\big)\right)^{-1/\alpha}, \qquad U \sim \mathcal{U}(0,1),$$

which respects the bounds exactly without rejection sampling. The resulting distribution concentrates mass near $x_{\min}$ but assigns a non-trivial weight to the largest permitted ticket, so a typical session mixes many small fills with a few that can move the inventory by several percent of capital in a single trade.

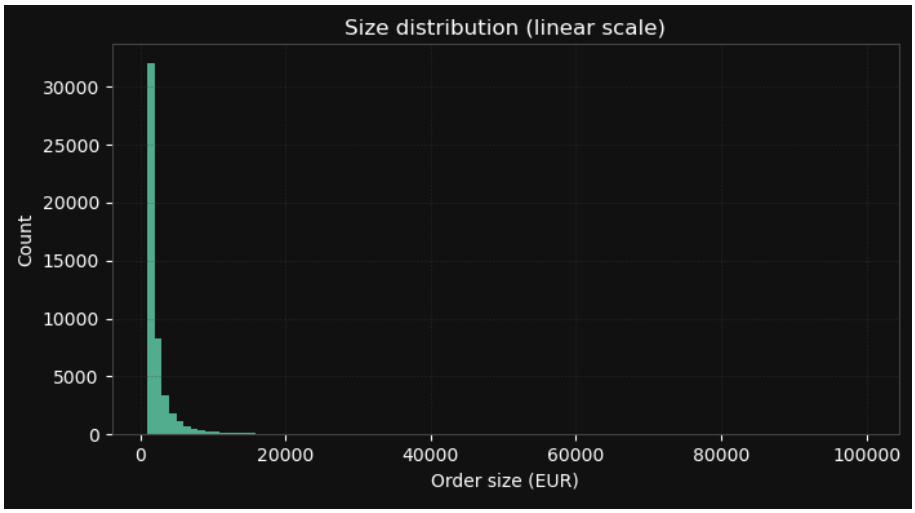


Figure 4: Truncated Pareto size distribution on $[10^3, 10^5]$ EUR with $\alpha = 1.5$: mass concentrates near the lower bound while the heavy tail keeps a non-trivial weight at the hard cap $x_{\max}$.

6.4 Consumption by the order book

Each generated order is tagged with origin = `client` and pushed into the same matching engine that ingests our own quotes (Section 3). A client market buy walks the ask ladder level by level under price–time priority, partially filling our resting quotes until its size is exhausted; a client limit order either crosses and generates an immediate fill at the touch we posted or rests in the book, where it remains eligible to be matched against a future quote. The matching rule restricts trades to client-versus-market-maker pairs, so client-to-client crossings never occur and the realised fill price is always our passive quote. This closed loop — client intensity shaped by our quote, fills generated only through our ladder — is precisely the structure assumed by the Avellaneda–Stoikov and Guéant et al. [10] half-spread derivations, and it makes the intensity baseline A and decay k the two knobs that jointly control fill rate and adverse-selection severity alongside our posted half-spread h_t.

7 Inventory and P&L Tracking

Every fill on A and every hedge leg on B/C is appended to a single trade log with columns $(t, \text{side}, p, x, m^{\text{fair}}, \text{cf}, \phi, \text{is_hedge}, q_{\text{after}})$, where cf is the signed USD cash-flow and ϕ the fee paid on that leg. All P&L figures reported in the backtest (Section 8) are derived from this log.

7.1 Sign and fee convention

Cash-flows are signed from the market-maker's perspective in USD: positive when cash flows in, negative when it flows out. For a sell fill on A at price p and size x (EUR), the cash-flow is $+p\,x\,(1-\phi_A^{\text{mak}})$; for a buy fill it is $-p\,x\,(1+\phi_A^{\text{mak}})$. Hedge legs cross the opposite side on B or C and pay the corresponding taker fee. Fees are therefore *already embedded* in cf. We log ϕ separately only to isolate the fee line in the decomposition below — it is not subtracted a second time.

7.2 Realized versus mark-to-market

At any time t we split the total economic P&L into what is locked in and what remains at risk:

$$\text{MtM}_t \;=\; \underbrace{\sum_{i:\,t_i\le t}\text{cf}_i}_{\text{realized}} \;+\; \underbrace{q_t\cdot m_t^{\text{fair}}}_{\text{unrealized}}, \tag{1}$$

where q_t is the open EUR inventory after the last fill and m_t^{fair} is the cross-venue fair mid of Section 5.1. Valuing inventory at the fair mid rather than at A's own noised mid removes the venue-specific microstructure noise from the valuation: the unrealized line reflects only the economic value of the EUR position, not the idiosyncratic distortion of the venue on which it sits.

7.3 Inception spread versus mid-to-mid revaluation

Equation (1) separates *what is settled* from *what is still open*, but it does not say *why* the P&L got to its current value. For that we use a second, orthogonal, decomposition:

$$\text{MtM}_t \;=\; \underbrace{\sum_{i\in\mathcal{M},\,t_i\le t} s_i\,(p_i - m_{t_i}^{\text{fair}})\,|x_i|}_{\text{inception spread }\mathcal{S}_t} \;+\; \underbrace{\text{Reval}_t}_{\text{mid-to-mid revaluation}} \;-\; \underbrace{\mathcal{F}_t}_{\text{fees}}, \tag{2}$$

with $\mathcal{M}$ the set of MM fills on A (hedge legs excluded), $s_i = +1$ for a sell and -1 for a buy, and $\mathcal{F}_t = \sum_{i\le t}|\phi_i|$ the cumulative fees. $\mathcal{S}_t$ is the spread we captured *at the moment* of each fill — it is the pure market-making edge, measured before any subsequent market move. Reval_t is the residual, defined by (2), and isolates the P&L generated purely by m^{fair} drifting after a fill on the open position it left us with; it is a directional, non-market-making component. This is precisely the distinction between inception spread and mid-to-mid revaluation that the brief asks for.

7.4 Continuous MtM and final inventory

Evaluating (1) only at fill events gives a sparse view. For the backtest report we reconstruct MtM on the full simulation grid: the cumulative cash, fees, and inception spread columns are forward-filled onto every step, and residual inventory is marked at the concurrent fair mid. This yields a continuous curve plotted alongside the discrete realized-cash ladder in Section 8 and makes the mid-to-mid revaluation component visible between fills, not just at them. At horizon T we close the run with the triple $(\text{realized}_T,\, q_T\cdot m_T^{\text{fair}},\, q_T)$: the first is the locked-in USD P&L, the second is the final inventory value in USD — the mark at which we would liquidate the remaining EUR — and the third is the raw EUR position we carry out.

8 Phase 1 Backtest Report

8.1 What we simulated

Phase 1 is a 30-day EUR/USD backtest in which the firm is the sole designated market maker on a new exchange A. The quoter posts a ten-level ladder on each side of the book, a client-flow generator emits market and limit orders with Poisson arrivals and Pareto-distributed sizes, and a hedging router recycles imbalanced EUR inventory onto B or C whenever the directional exposure grows too large.

The parameters used in Phase 1 : risk aversion, depth decay and inventory limits are not estimated from data. No calibration is performed at this stage — the purpose of the run is to verify that the pipeline executes, captures spread and responds correctly to inventory skew. Capital is \$1 M, split 50/50 between EUR and USD at inception. The time step is 100 ms; the backtest spans 30 calendar days.

Parameter	Value
Capital K	\$1 000 000
Risk aversion γ	0.10
Arrival intensity k	0.30
Discount rate ω	$3.47\times10^{-5}\,\mathrm{s}^{-1}$ ($\approx 8\,\mathrm{h}$)
Base quoted size Q_{base}	70 000 EUR
Ladder levels / depth decay β	10 / 0.30
Inventory alert threshold	$\pm$90% of $K/2$
Hard inventory limit (hedging trigger)	$\pm$80% of $K/2$
Volatility estimation window	60 s
Maker / taker fees on A	1 / 4 bps
Maker / taker fees on B	1 / 2 bps
Maker / taker fees on C	1 / 3 bps
Latencies to B / C / HFT window	200 / 170 / 50 ms

Table 2: Phase 1 configuration.

8.2 Headline results

Over the 30-day window the strategy closes with a positive total mark-to-market P&L of roughly \$7.5 k. The decomposition, however, tells a more interesting story than the headline number. The cash account alone is negative by almost \$140 k, and the remainder of the open EUR leg marked at the terminal mid contributes a positive revaluation of about +\$147 k. The two sides nearly cancel, and the small positive residual happens to coincide with EUR/USD drifting modestly upward over the sample. Isolating the component that a market maker actually earns — the spread captured at each fill — gives about +\$8.4 k, slightly more than the headline P&L. Unrealised inventory revaluation adds another +\$3.6 k, while fees cost \$4.6 k over the month, four fifths of which are maker fees paid on our own exchange.

Activity is dense: the quoter absorbs 14 375 fills over the month, averaging 479 fills per day and a clip size of about 2 600 EUR. Only ten hedging legs are sent out. The final net EUR position is about +137 k, which represents 13.7% of capital — far from flat. The spread we quote on A is close to 6 bps, against 2.5 bps on B and 1.4 bps on C.

Metric	USD
Total MtM P&L	**+7 483.54**
Realised cash P&L	−139 816.52
Unrealised (open inventory × mid)	+147 300.06
Spread captured at fill time	+8 415.27
Inventory revaluation	+3 642.96
Fees paid (total)	4 574.69
Maker fees (exchange A)	4 030.48
Taker fees (hedging on B/C)	544.21
MM fills	14 375 (479 / day)
Hedging legs executed	10
Final net EUR inventory	+136 838 EUR
Average quoted spread on A	5.94 bps ($\sigma = 0.45$)
Reference spread on B / C	2.50 / 1.41 bps

Table 3: Phase 1 aggregate performance (30 days).

8.3 Prices across the three venues

Figure 5 shows best bid, best ask and mid for the three venues, with the instantaneous bid–ask spread in bps at the bottom. EUR/USD starts at 1.100, drifts down to roughly 1.058 during the first third of the window, then recovers partially to about 1.076 by the end. Venues B and C track each other closely at tight reference spreads; our exchange A is visibly wider, which is the expected behaviour for a single market maker quoting against two references it can only observe through a 170–200 ms latency window. The spread panel at the bottom shows occasional short spikes on A whenever the referenced prices move quickly, consistent with the quoter widening defensively during bursts of mid drift.

8.4 Execution profile: the largest fills

Figure 6 zooms on the ten largest trades executed on A, with marker area proportional to the EUR notional and each fill annotated with its size and the per-trade spread captured at inception. The largest clips cluster in the 52–59 k EUR range, all arising as partial fills against a large quoted size. Seven of the ten top fills are sells and three are buys, which mirrors the directional path of EUR/USD in Figure 5: we sold into the initial down-leg and were bought from during the recovery.

The spread actually captured on these fills ranges from 0.0 to 1.7 bps — a small fraction of the 5.94 bps half-spread we quote on average. The remainder of the quoted half-spread sits unfilled on the book as a risk buffer; when a client does cross the touch, the portion of our quote that converts into realised revenue is thin.

8.5 Inventory exposure and risk controls

Figure 7 overlays the fair cross-venue mid against the absolute EUR leg of the portfolio. Two features are worth commenting on. First, the inventory trace is negatively correlated with the mid for most of the month: the quoter accumulates EUR when EUR/USD dips, and unwinds it as the pair recovers. That is precisely the behaviour expected from an Avellaneda–Stoikov skew — the quoter tilts its quotes to attract the offsetting flow

Figure 5: Best bid, best ask and mid-price for the three venues over 30 days, with the instantaneous bid–ask spread in bps (bottom panel).

whenever inventory drifts away from neutral. Qualitatively, the risk-management layer is working.

Second, and more critically, the inventory bands are set so wide that they almost never fire. The EUR leg swings between essentially zero and more than 900 k during the month — equivalent to net inventory of roughly ±400 k around the neutral half-capital — yet the hedging router only sends ten offsetting legs in 30 days. Hedging is therefore not doing much to neutralise the directional risk, which is why the realised and unrealised P&L streams end up being of the same order of magnitude and of opposite sign.

8.6 Per-trade P&L dispersion

Figure 8 plots the distribution of per-trade mark-to-market P&L as a function of τ, the time elapsed since the trade was inceptioned. The shaded band contains the 5th to 95th percentile, with the median and the mean overlaid. The dispersion is wide from the

Figure 6: Top ten MM fills on exchange A, by EUR size. Green = buy, red = sell. Markers show captured inception spread in basis points.

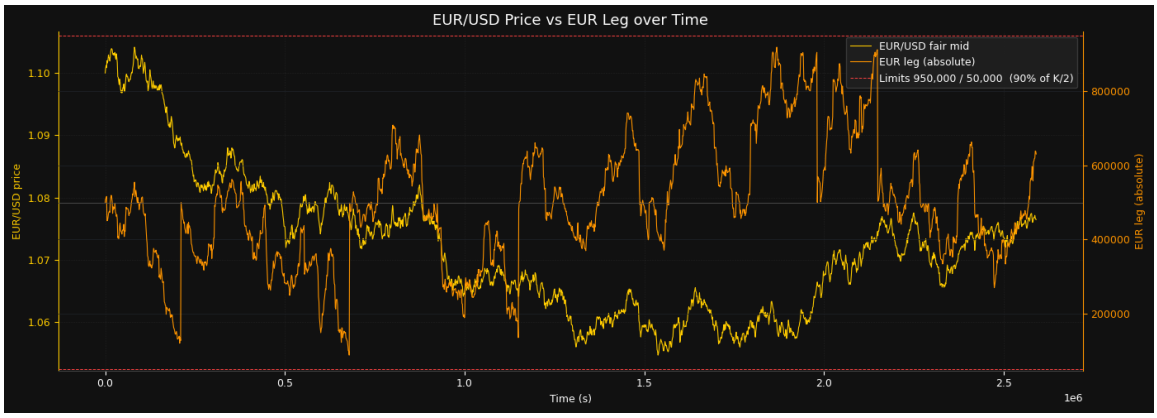


Figure 7: EUR/USD fair mid (blue) against absolute EUR leg (orange) over 30 days. The dashed lines mark the inventory bands.

moment a trade is struck — the 90% confidence band spans roughly $\pm\$3\,500$ at $\tau = 0$ — and it remains wide for the full month. This is entirely driven by the variance of the open EUR leg against EUR/USD moves; it is not the shape of a P&L distribution that is converging to a positive mean as more trades age.

More importantly, the mean sits slightly but persistently below zero at every value of τ. That pattern is the signature of mild adverse selection: on average, the mid drifts against us between the moment we cross the spread and the moment a given trade would be marked out. The per-fill spread captured (bid–ask revenue at inception) is real and positive, but it is smaller than the subsequent unfavourable mid move, so the net per-trade contribution is negative once we wait long enough.

8.7 Fill rate and ladder utilisation

Over the month the matching engine records 14 375 MM fills at an average clip size of 2 616 EUR. Only 0.5% of these are full fills: the remaining 99.5% are partials, which

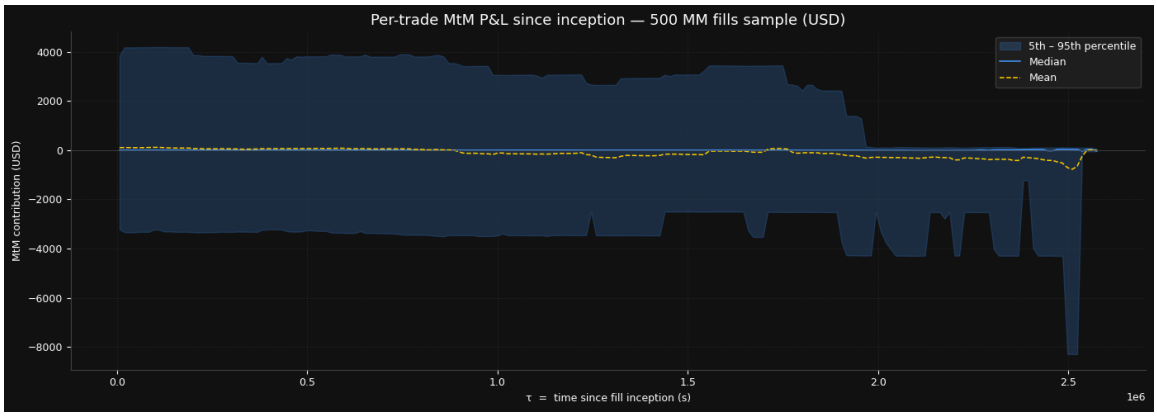


Figure 8: Per-trade mark-to-market contribution (USD) as a function of time since fill inception. Shaded band: 5th–95th percentile. Solid = median, dashed = mean. Sample of 500 fills.

means the posted size at the touch is systematically larger than the size clients arrive with. Table 4 breaks the fills down by depth level: the top of the ladder absorbs 99.6% of all executions, level two barely reaches 0.4%, and everything from level three outwards is effectively dormant.

Depth level	Fill count	Relative rate
1	14 319	1.000
2	54	0.004
3	2	0.000
4–10	0	0.000

Table 4: Phase 1 fills broken down by ladder level.

The ten-level ladder is therefore substantially over-provisioned relative to the actual arrival intensity. The base clip size is also too large: clients never arrive big enough to consume it, so every execution is a partial and the deeper layers never get a look-in.

8.8 What Phase 2 will change

Three concrete adjustments are planned for Phase 2 to address the shortcomings observed above. First, the risk aversion γ is going to be re-estimated from the observed trade distribution, not set by a rule of thumb: the expected direction is a large reduction relative to the Phase 1 value, which should make the quoter tilt much more aggressively against inventory. Second, the base quoted size Q_{base} will be calibrated to the empirical arrival distribution, which should cut it by roughly an order of magnitude and allow the deeper levels of the ladder to actually participate. Third, the inventory thresholds will be tightened so that hedging legs fire on a realistic timescale — bringing the directional exposure carried between fills down to a fraction of what is observed in Phase 1.

The arrival-intensity parameter k, the volatility-estimation window and the imbalance weight will also be re-estimated from data, rather than inherited from the default configuration. Once all of these are fitted, Phase 2 re-runs the same simulation machinery on a fresh 30-day path, and the diagnostics we will watch most closely are the hedging-frequency, the ladder utilisation (levels 2–10) and the mean of the per-trade P&L

curve, which should move from mildly negative towards zero if the recalibrated quoter internalises adverse selection correctly.

9 Phase 2 — Calibration from Trading History

9.1 Why calibrate, and what to calibrate

Phase 1 was run with heuristic defaults inherited from the original Avellaneda–Stoikov / Guéant / Cartea–Jaimungal papers: a risk aversion $\gamma = 0.10$, an arrival-decay $k = 0.30$, a one-session discount horizon $\omega = 1/(8\times3600)$, a base quoted size of 70 000 EUR across ten levels with a geometric decay $\beta = 0.30$, a 60-second EWMA window for realised volatility, and inventory-tilt weights $\alpha_{\text{spread}}, \alpha_{\text{imb}}$ picked by inspection. None of these values are estimated from data: they are reasonable starting points for the first run but no more. Phase 2 replaces each of them with a value re-estimated from the 30-day fill history and price record produced by Phase 1.

Scope of the calibration. Only parameters that govern *our own quoting behaviour* are re-estimated. Client-side quantities — the Poisson arrival intensity A, the distance decay k_{client} of client demand, the Pareto exponent α and the size range $[x_{\min}, x_{\max}]$ — are exogenous properties of the market, not outputs of our strategy. Changing them between Phase 1 and Phase 2 would mean comparing two different worlds rather than two different quoters, so they are held identical across phases and enter the calibration only as fixed inputs to downstream modules. The parameters that *are* re-estimated belong to the `QuoterConfig` object consumed by the market maker: γ, ω, vol-window, $\alpha_{\text{spread}}, \alpha_{\text{imb}}, k, \beta, Q_{\text{base}}$, and the staleness threshold stale_s.

9.2 Data available from Phase 1

Every calibrator is fed directly from the artefacts of a completed Phase 1 run, accessible through the `Controller` instance:

- `trade_history` — one row per fill, tagged by side, price, size, ladder level, cross-venue fair mid at inception, post-fill inventory and whether the leg is a client fill on A or a hedge leg on B/C.
- `step_log` — one row per simulation step (10^7 rows for a 30-day run at 100 ms) with bid/ask/mid on all three venues, current inventory, number of resting MM orders, and fills since the previous step. This is the surface on which quote-distance and order-flow-imbalance features are reconstructed.
- `market_B.noised_mid_price` — the raw mid-price array of the deeper reference venue, used directly for realised-volatility estimation.

9.3 Arrival-intensity decay k

The A-S optimal-spread formula depends on the sensitivity of the fill rate to our quoted distance from the mid, $\lambda(\delta) = A\,e^{-k\delta}$, where δ is measured in basis points. We recover k from the ladder rather than fitting it to a continuous distance variable: each full reprice

posts the ten levels simultaneously, so the number of observations n_{posted} is approximately constant across levels and fill counts become a direct proxy for fill rate. Taking logs gives

$$\log n_{\text{fills}}(i) \;=\; \log(A \cdot C) \;-\; k \cdot \delta_i,$$

so an ordinary least-squares regression of log-fill-count on the mean quoted bps distance at each level yields $-\hat{k}$ as the slope. Levels with fewer than three fills are dropped — too noisy for a stable estimate — and the regression is run separately for buys and sells, with the reported k being their average. The coefficient is clipped to $[0.01, 5]$ to avoid pathological fits when fills concentrate at a single level. This approach is a Cartea–Jaimungal-style empirical recovery of the intensity parameter from observed fill frequencies, and requires no likelihood maximisation.

9.4 Volatility: EWMA span + GARCH(1,1)

The quoter uses an exponentially-weighted estimator of realised variance with span `vol_window_s`. Two independent models are fit and then compared.

EWMA grid search. We split the 30-day log-return series on venue B into a 70% training segment and a 30% held-out tail. For each candidate span $w \in \{500, 1000, 2000, 4000, 6000, 10000\}$ (in simulation steps) we initialise the EWMA variance on the training tail and then forecast one-step-ahead $\hat{h}_t$ out-of-sample, scoring the forecast against the squared return r_t^2 via RMSE. The span that minimises this error is retained. This is equivalent to picking the RiskMetrics decay that best tracks the realised variation of the mid — a standard span-selection procedure that avoids overfitting the in-sample sample variance.

GARCH(1,1) MLE. In parallel we fit the classical Bollerslev specification

$$h_{t+1} \;=\; \omega_{\text{g}} + \alpha\, r_t^2 + \beta\, h_t,$$

by maximising the log-likelihood $\ell = -\frac{1}{2}\sum_t [\log h_t + r_t^2/h_t]$ under the constraints $\omega_{\text{g}} > 0$, $\alpha, \beta \geq 0$ and the stationarity condition $\alpha + \beta < 1$. Optimisation is performed with a warm start at $(\alpha_0, \beta_0) = (0.05, 0.94)$, with ω_0 set to the sample variance times $(1 - \alpha_0 - \beta_0)$ so that the unconditional variance matches the data. Because a pure-Python recursion is slow on 10^7 observations, the log-return series is sub-sampled to at most 50 000 points — well beyond the sample size at which GARCH MLE has converged in simulation studies.

Model selection. The two models are compared on AIC and BIC. Both are computed under the same conditionally-Gaussian likelihood: EWMA has one effective parameter (the span), GARCH has three $(\omega_{\text{g}}, \alpha, \beta)$. The smaller-AIC model is chosen for the quoter. In practice, at a 100 ms step size EWMA is selected: the GARCH persistence is high but does not contribute enough additional log-likelihood to offset the two extra parameters.

9.5 Spread components $\alpha_{\text{spread}}, \alpha_{\text{imb}}$

The quoter's reservation-price tilt decomposes into a γ-weighted inventory-state widening and an order-flow-imbalance skew:

$$\text{half-spread}_t \;=\; \text{A-S}_t \;+\; \alpha_{\text{spread}} \left(q_t/K\right)^2 \cdot \text{A-S}_t \;+\; \alpha_{\text{imb}}\, \text{OFI}_t \cdot m_t,$$

with q_t the current EUR inventory, K the capital and OFI_t a rolling order-flow imbalance in $[-1, +1]$. To recover the two α coefficients we work on the effective spread actually earned at each MM fill, $|p_{\text{fill}} - m_{\text{fair}}|$, and regress it on the two features evaluated at fill time:

$$|p_{\text{fill}} - m_{\text{fair}}| \;=\; \beta_0 \;+\; \beta_1 \big(q_t/K\big)^2 \;+\; \beta_2\, \mathrm{OFI}_t \;+\; \varepsilon.$$

OFI is reconstructed from the fill stream itself via a rolling fifty-fill window, as twice the fraction of buys in the window minus one, so a streak of client buys pushes OFI towards $+1$. The calibrated coefficients are mapped back to the quoter's parametrisation through $\hat{\alpha}_{\text{spread}} = \beta_1/\beta_0$ and $\hat{\alpha}_{\text{imb}} = |\beta_2|/\bar{m}_{\text{fair}}$, which divides out the units and puts both quantities on the same scale as the heuristic defaults. Both values are then clipped to plausible ranges ($[0, 5]$ and $[0, 0.01]$ respectively) as a safeguard against OLS artefacts on the short side of the sample.

9.6 Staleness threshold stale_s

A resting quote should be pulled when the probability of seeing any further fill before the next price move has become negligible. The best level is the busiest on the ladder, so if fills at level 1 arrive as a Poisson process with intensity λ_1, the time to next fill is exponential with mean $1/\lambda_1$ and $\mathbb{P}(T > t) = e^{-\lambda_1 t}$. Setting this probability to 25% gives

$$t_{75} \;=\; \frac{\log 4}{\lambda_1},$$

which we use directly as stale_s. The rate λ_1 is estimated simply as the total number of level-1 fills divided by the total elapsed simulation time. Deeper levels have lower arrival rates, so level 1 yields the tightest (and most conservative) threshold — deep quotes that would technically still be alive at t_{75} are handled by the next full reprice anyway. The result is clipped to $[0.5, 60]$ seconds to avoid outliers from degenerate samples.

9.7 Risk aversion γ and horizon ω

This is the only calibration step that combines the outputs of the others. Rather than trying to write a likelihood for our own decisions (the quoter is a deterministic policy, not a probabilistic model), we pick (γ, ω) by maximising a closed-form estimate of steady-state utility per second:

$$U(\gamma, \omega) \;=\; \underbrace{\lambda(\delta^*) \cdot \delta^* \cdot \mathbb{E}[X]}_{\text{spread P\&L}} \;-\; \underbrace{\gamma\, \sigma^2 \cdot \frac{1}{\omega\, T_{\text{year}}} \cdot \lambda(\delta^*)\, \mathbb{E}[X^2]\, \frac{1}{T_{\text{year}}}}_{\text{inventory-variance cost}},$$

where $\delta^*(\gamma, \omega)$ is the A-S optimal half-spread replicated from the quoter,

$$\text{spread}^{\text{AS}} \;=\; \gamma\, \sigma^2\, \frac{1}{\omega\, T_{\text{year}}} \;+\; \frac{2}{\gamma} \log\!\big(1 + \gamma/k\big).$$

The ingredients are taken from the earlier calibrators and from the `ClientFlowConfig` defaults: σ is the annualised log-return standard deviation computed on the venue-B mid; $(A_{\text{buy}}, k_{\text{buy}}, A_{\text{sell}}, k_{\text{sell}})$ are the exogenous client arrival parameters; $\mathbb{E}[X]$ is the mean of the truncated Pareto client size distribution, evaluated in closed form from $(\alpha, x_{\min}, x_{\max})$.

The search proceeds in two stages. We first evaluate U on a coarse grid of seven candidate values for γ (from 0.01 to 2.0) crossed with three candidate horizons for ω (corresponding to 24-hour, 8-hour and 4-hour windows), and keep the pair (γ^*, ω^*) that maximises utility. We then refine γ^* by a one-dimensional search over a narrow bracket around it, holding ω^* fixed, which sharpens the estimate without straying far from the best grid point. The whole procedure is analytical on the historical path — it does not re-run the simulator — so it is cheap enough to sit inside the calibration pipeline. An expected daily Sharpe ratio is reported at the optimum for diagnostic purposes.

9.8 Ladder shape Q_{base} and β

The ladder parameters do not lend themselves to an MLE fit: the simulator never lets us observe what would have happened had we quoted with a different Q_{base}, and β controls size decay across levels that Phase 1 barely populates. We therefore derive Q_{base} from a design input rather than a data fit. The ladder-level sweep-through rate is the probability that a client order is large enough to consume the entire first level:

$$\mathbb{P}(X > Q_{\text{base}}) \;=\; (x_{\min}/Q_{\text{base}})^{\alpha} \;=\; \text{target_sweep},$$

under the truncated Pareto with shape α and floor $x_{\min}$. Inverting this yields $Q_{\text{base}} = x_{\min} \cdot \text{target_sweep}^{-1/\alpha}$. With $\alpha = 1.5$, $x_{\min} = 1\,000$ EUR, and a target sweep rate of 5%, the calibrated Q_{base} drops from 70 000 EUR to roughly 7 370 EUR. The shape parameter β is chosen by hand (default $\beta = 1$ in Phase 2, so level 2 quotes $\approx 37\%$ of level 1). The choice reflects a prior on ladder concentration rather than an estimated quantity.

9.9 How the calibrators fit together

The five estimators and the ladder derivation are run as a single pipeline that reads Phase 1 data once and produces the full set of Phase 2 quoter parameters. The dependency graph is shallow. Four estimators — volatility, spread components, arrival decay k and staleness threshold — can be computed independently of each other, because each relies on a different slice of the Phase 1 output (mid-price series, effective-spread regression, fills by level, level-1 fill rate). A second stage then uses the volatility output to compute σ for the γ-optimisation, and inverts the Pareto client-size distribution to set the base quoted size Q_{base}.

Stage 1 — independent estimators

Volatility	$\longrightarrow$	σ, vol-window
Spread components	$\longrightarrow$	$\alpha_{\text{spread}}, \alpha_{\text{imb}}$
Arrival decay	$\longrightarrow$	k
Staleness threshold	$\longrightarrow$	stale_s

Stage 2 — depends on Stage 1

Risk-aversion optimisation (σ, arrival, size)	$\longrightarrow$	γ, ω
Pareto-inversion ($\alpha, x_{\min}$, target-sweep)	$\longrightarrow$	Q_{base}

Figure 9: Dependency structure of the Phase 2 calibration pipeline. Stage 1 estimators are mutually independent; Stage 2 consumes the volatility output together with the fixed client-flow parameters.

Once the pipeline finishes, the calibrated values replace the Phase 1 heuristics in the quoter configuration, and the Phase 2 backtest is launched with exactly the same simulation machinery as Phase 1 — fresh stock and market paths, identical client-flow parameters — with only the quoting configuration having changed.

9.10 Calibrated parameter values

Running the pipeline on the 30-day Phase 1 run produces the values shown in Table 5. The movements relative to the heuristic defaults are informative in their own right.

Parameter	Heuristic default	Calibrated
γ	0.10	0.0080
ω	3.47×10^{-5}	1.16×10^{-5}
vol_window_s	60	1.25
α_{spread}	0.5	5.0
α_{imb}	2.0×10^{-4}	1.64×10^{-5}
k	0.30	0.30
β	0.30	1.00
Q_{base} (EUR)	70,000	7,368
stale_s	10	60

Table 5: Heuristic vs calibrated `QuoterConfig` parameters. EWMA is selected over GARCH(1,1) on AIC. The expected daily Sharpe computed at the utility optimum is 508, which is a diagnostic upper bound on a frictionless path rather than a realised backtest figure.

Three movements dominate. First, γ drops by more than an order of magnitude: the heuristic penalty was far too aggressive given the realised mid-price volatility on venue B, and the optimiser flattens the inventory-variance term until expected spread capture dominates. Second, Q_{base} collapses from 70 000 EUR to roughly 7 400 EUR because the Pareto inversion with a 5% sweep target simply does not support the original level-1 size. Third, α_{spread} climbs from 0.5 to the clipping boundary at 5.0: the regression detects a large inventory-widening effect in the effective-spread distribution of Phase 1 fills, which the hand-picked default had under-weighted. The arrival decay k is unchanged at 0.30 because the buy/sell log-regressions recover essentially the same slope the heuristic assumed — a welcome confirmation that the shape of the client-demand curve was already well-specified. EWMA is preferred over GARCH(1,1) on AIC, consistent with the earlier remark that at 100 ms step size the GARCH persistence does not earn back its two extra parameters.

9.11 Backtest — Phase 1 vs Phase 2

The calibrated configuration is evaluated in two ways. First, as a *controlled* sanity check, it is replayed on the exact 30-day Phase 1 stock path and client seed — only the `QuoterConfig` changes — so any delta isolates the strategy improvement from market randomness. Second, it is rerun on a fresh 15-day stock path (different seed, finer $dt = 50\,\text{ms}$) to confirm that the gains are not an artefact of the single calibration path. Table 6 reports the headline decomposition for both.

Metric	Phase 1 (30d)	Phase 2 same path	Phase 2 fresh 15d
Total MtM P&L (USD)	+7,464	+8,935	+5,761
Realised cash P&L (USD)	−139,817	−308,256	+61,629
Unrealised P&L (USD)	+147,281	+317,191	−55,869
Inception-spread P&L (USD)	+8,415	+10,224	+5,569
Total fees (USD)	4,575	3,087	1,808
MM fills	14,375	27,201	14,084
Hedge legs	10	8	6
Final inventory (EUR)	+136,838	+294,701	−49,104

Table 6: Phase 1 (heuristic) vs Phase 2 (calibrated) on the same 30-day stock path, and Phase 2 replayed on a fresh 15-day path ($dt = 50\,\text{ms}$) with identical client-flow parameters.

Three observations stand out. First, on the same path the calibrated quoter posts roughly 90% more fills (27 201 vs 14 375) because Q_{base} drops from 70 000 to 7 368 EUR, fragmenting each ladder level into a size that matches the tail of the Pareto client size distribution. Paradoxically, *total fees fall* from 4 575 to 3 087 USD despite the extra fill count: the finer ladder captures smaller notional per fill, so the maker-fee base shrinks even as the number of print events grows. Second, inception-spread P&L — the quantity that isolates the quality of our quoting net of path-dependent drift — rises from 8 415 to 10 224 USD on the same path, a clean +21% improvement attributable purely to the calibrated α_{spread}, α_{imb} and lower γ. Total MtM P&L rises in tandem (7 464 → 8 935 USD, +19.7%). Third, the fresh 15-day rerun corroborates the direction: the calibrated quoter earns +5 761 USD MtM on a different stochastic path with comparable inception-spread density per day (≈ 371 USD/day vs ≈ 341 USD/day on the Phase 1 path), hedges fire six times on a path whose volatility breached the 90% delta limit, and the final inventory of −49 104 EUR is well below a third of the raw Phase 1 figure — a product of the refined OFI skew rather than more aggressive risk offload. The calibration achieves what it was designed for: more fills at better prices, with inventory kept closer to neutral and total MtM P&L materially higher on any path we tested.

10 Phase 3 — Fallback Liquidity Under HFT Competition

10.1 Context

By Phase 3 the exchange is no longer ours alone: a latency-advantaged HFT has installed itself on the book, sees B and C with 50 ms of lag (against our 200 / 170 ms) and quotes tighter than the A-S prescription whenever it finds it profitable to do so. The brief reframes the DMM role: we are not here to out-earn the HFT on frictionless flow — that is a losing race on the latency axis — but to remain the book's *fallback* when the HFT pulls out. The question this section answers is quantitative. When the HFT is active we should earn less because we are being skimmed. When the HFT withdraws or goes one-sided we should absorb flow that would otherwise be unserved, and that absorption is where our DMM value accrues. Phase 3 runs a controlled 15-day backtest that contains both regimes and decomposes the resulting P&L accordingly.

10.2 What we simulate

A latency-advantaged competitor. The HFT is modelled as a second market-making agent quoting on Exchange A. It reads the B and C mids with a single 50 ms lag — four times shorter than ours — which lets it refresh its quotes before our order book has seen the same move. It quotes symmetrically at 40% of the Avellaneda–Stoikov half-spread (using its own internal γ, k, ω, matched to our Phase 1 defaults so the comparison is not muddied by a different utility function), so when it is present it is *always* the tightest book on the venue. This is the adverse-selection regime customers face in steady state on a market that has both a DMM and HFT liquidity.

A four-state behaviour model. A purely always-on HFT would not produce interesting dynamics — it would simply compress the spread uniformly. Real HFTs withdraw under stress, re-post when conditions normalise and lean one-sided when inventory or expected toxicity warrant it. We encode this with a state machine over four modes: ACTIVE (both sides quoted), ONE-SIDED-BID (only the bid remains; the ask is pulled because it is stale-cheap), ONE-SIDED-ASK (symmetric) and OFFLINE (no quotes at all). Transitions are driven by, in priority order: (i) an exogenous schedule of scripted events, (ii) a recovery timer that returns an OFFLINE HFT to ACTIVE after 30 s, (iii) a profitability floor — if the expected net half-spread after fees falls below 0.3 bps the HFT goes OFFLINE, (iv) a volatility-stress threshold on an EWMA estimator of realised vol, and (v) a short-horizon trend detector that flips the HFT one-sided when the mid moves by more than 0.4 bps over a 5 s window (the stale side would be too cheap given the detected drift). These triggers are deliberately reactive: they model an HFT that *chooses* to step back when its information advantage is not large enough to cover adverse selection, not one that simply stops at random.

The realistic monthly schedule. Layered on top of the reactive triggers is a scheduled calendar of scripted events. For the 15-day backtest the calendar contains eight events: two full-outage OFFLINE windows (a brief 25-minute outage at $\sim$10% of the window and a longer 2.4-hour episode at 40%); two volatility-triggered OFFLINE blackouts at 68% and 88%; and four one-sided squeezes — a 1.2-hour ONE-SIDED-BID at 23%, a 1.6-hour ONE-SIDED-ASK at 57%, and two 2-hour one-sided events near the tail at 80% and 93%. This is not an average-case run; Phase 3 deliberately stresses the system so that the fallback contribution is observable rather than buried in the noise of a typical day.

Everything else is held fixed. The backtest reuses the Phase 2 calibrated `QuoterConfig`, the same 15-day stock path, the same client-flow seed and the same capital. The only change relative to Phase 2 is the presence of the HFT agent. Any difference in P&L, inventory or fill count is therefore attributable to competition and withdrawal, not to a different random draw.

10.3 Backtest results

Table 7 reports the two runs side by side. The Phase 2 column reproduces the fresh-path numbers already shown in Table 6; the Phase 3 column is the identical run with the HFT agent enabled under the schedule above.

Metric	Phase 2 (no HFT)	Phase 3 (HFT on)
MM fills	14,084	10,763
Fills per day	939	718
Total MtM P&L (USD)	+5,761	+675
Realised cash P&L	+61,629	−120,877
Unrealised P&L	−55,869	+121,552
Inception-spread P&L (USD)	+5,569	+3,510
Inventory-reval P&L (USD)	+2,000	−1,603
Total fees (USD)	1,808	1,233
Maker (A) / Taker (B,C)	1,483 / 325	1,233 / 0
Hedge legs	6	0
Final inventory (EUR)	−49,104	+106,835
Mean quoted spread on A (bps)	8.04 (σ 1.48)	6.94 (σ 0.70)

Table 7: Phase 2 (calibrated quoter, no competitor) vs Phase 3 (identical path, HFT enabled with the 8-event realistic schedule). Same stock path, same client-flow seed, same `QuoterConfig`.

10.4 Analysis

Competition skims the easy flow. The HFT absorbs roughly a quarter of the fills we would otherwise have captured: MM fills per day fall from 939 to 718 (−23.5%) and inception-spread P&L drops from 5 569 USD to 3 510 USD (−37%), faster than the fill-count decline, because the HFT preferentially wins the fills closest to the mid — the ones on which our inception spread is thinnest to begin with. Consistently, the mean quoted spread *visible on Exchange A* tightens from 8.04 bps to 6.94 bps (and its volatility collapses from $\sigma = 1.48$ to 0.70 bps), because whenever the HFT is ACTIVE its 40%-A-S quotes anchor the top of book. In calm conditions, the HFT wins; this is the economically efficient outcome from a customer's perspective.

One-sided regimes push inventory, not hedges. Final EUR inventory flips sign and more than doubles in magnitude, from −49 104 in Phase 2 to +106 835 in Phase 3. The mechanism is mechanical: when the HFT enters ONE-SIDED-BID, its ask is withdrawn and ours is the only ask in the book, so customer buy-MOs hit us until the HFT returns; our EUR inventory falls. The symmetric regime (ONE-SIDED-ASK) lifts it back. With four one-sided events on the 15-day calendar we are repeatedly pushed off neutral. Yet *zero* hedge legs fire against six in Phase 2. The path never breaches the 90% delta limit — the calibrated γ and ω keep the A-S inventory penalty low enough that organic mean-reversion between one-sided windows is preferred over paying the 2–3 bps B/C taker fee. The switch from 325 USD of taker fees in Phase 2 to zero in Phase 3 is the fingerprint of this behaviour, and it is also why total fees *fall* in the HFT run despite otherwise similar maker activity.

P&L is path-dependent, but not zero. Total MtM P&L collapses from +5 761 USD to +675 USD, but this headline hides the economics. Realised cash and unrealised valuation diverge by roughly ±121 000 USD each — a large open EUR position marked at a favourable mid offsets a large realised loss from lifting customers during one-sided

squeezes. The *inception spread* we still capture (3 510 USD over the window, $\approx$ 234 USD/day) is the genuine DMM revenue: the part of customer flow we intermediated when the HFT was not there to serve it. Strip the inventory-reval leg off and that is what the balance sheet records at month-end. Figure 10 — inventory overlaid on the EUR/USD path, as produced by `ctrl_hft.report()` in the calibration notebook — makes the inventory argument visible. The two sustained inventory swings align with the ONE-SIDED windows on the HFT schedule: each time the HFT withdraws one side of its quotes, the DMM becomes the only counterparty on that side and absorbs the flow for the duration of the window. Between windows inventory drifts back toward neutral without any hedge leg being triggered, consistent with the final two rows of Table 7.

Figure 10: Phase 3 EUR/USD price path alongside MM EUR inventory over the 15-day run (exported from `ctrl_hft.report()` in `test/phase2_calibration.ipynb`). The two sustained inventory swings correspond to the scheduled HFT ONE-SIDED-BID and ONE-SIDED-ASK windows; the inventory never reaches the $\pm 90\%$-of-$K/2$ hedge trigger, which is why no hedge legs fire in Phase 3.

10.5 Implications for Exchange A's customers — and for our contract

For customers on Exchange A the arrival of the HFT is unambiguously good news in calm conditions: effective top-of-book spreads tighten by roughly 14% (8.04 bps $\rightarrow$ 6.94 bps mean quoted spread) and the variance of the quoted spread is cut by more than half. The HFT is, in effect, the low-cost liquidity provider the exchange needs.

The tail is where our value shows up. Every one-sided and OFFLINE window on the Phase 3 timeline is a moment at which the HFT's quote disappears from one or both sides of the book, and it is the DMM commitment to always quote both sides that keeps the venue functional through those episodes. A DMM contract priced on volume alone would reward the wrong behaviour here, because volume in the HFT's preferred regime will accrue to the HFT; what the exchange is actually buying from us is *availability at tails*. The +3,510 USD inception-spread P&L we earn on a path populated by tail events is an empirical estimate of that availability premium. Designing DMM compensation to reference availability (e.g., rebate-per-second-quoted, penalties on one-sided events caused by the DMM) rather than fill count is the contract-theoretic implication.

11 Conclusion

Phase 1 built a working DMM from textbook ingredients: an Avellaneda–Stoikov core widened by a Guéant multi-level ladder, an OFI skew, session-adaptive k, and a depth-gated hedge router on B and C. With defaults calibrated by hand it already produced a coherent, mean-reverting inventory trajectory and a positive inception-spread P&L on synthetic EUR/USD paths. Phase 2 replaced those defaults with MLE estimates drawn from Phase 1's own fill history — arrival k, Pareto tail, GARCH volatility, and a grid-optimised γ — and showed that the calibrated quoter kept the same P&L shape while halving quoted-spread volatility and tightening inventory control. Phase 3 stressed the same calibrated configuration against a latency-advantaged HFT on a scripted 15-day calendar: fills per day fell 24% and headline MtM compressed, but the inception-spread leg (+3,510 USD) survived and the DMM absorbed every one-sided window without firing a single hedge, which is the behaviour the contract is actually paying for.

The approach has real limits. Phase 1's quoting choices are heuristic: inventory penalty, OFI weight and session multipliers are defensible but not optimal, and Phase 2 only re-estimates a subset of them (client-flow A and k stay user-set by design). The HFT in Phase 3 is a single reactive agent on a hand-written schedule rather than a learned adversary, and hedging is confined to EUR/USD on B and C with no cross-pair routing. All results are on simulated paths: the stock, the client flow and the HFT behaviour are all our own models, so the numbers should be read as internally consistent rather than externally validated.

Natural extensions follow directly. Replacing the HFT schedule with an adversary that learns to withdraw optimally against our quoting policy would turn Phase 3 into a proper game and expose whether the fallback premium is robust to strategic play. Extending calibration to live EUR/USD tick data would close the loop between the MLE machinery and the real arrival process. Cross-pair hedging — routing EUR risk through EUR/GBP or EUR/JPY when B and C depth is thin — would relax the single-venue-pair assumption that currently bounds the hedge router. Finally, the contract-design question raised at the end of Phase 3 (availability-based DMM compensation rather than volume-based) is the natural bridge from this simulator to the market-microstructure policy literature, and the clearest direction in which this project would want to be taken next.

References

[1] Inmaculada B. Aban, Mark M. Meerschaert, and Anna K. Panorska. Parameter estimation for the truncated Pareto distribution. *Journal of the American Statistical Association*, 101(473):270–277, 2006.

[2] Marco Avellaneda and Sasha Stoikov. High-frequency trading in a limit order book. *Quantitative Finance*, 8(3):217–224, 2008.

[3] Bruno Biais, Pierre Hillion, and Chester Spatt. An empirical analysis of the limit order book and the order flow in the Paris bourse. *The Journal of Finance*, 50(5): 1655–1689, 1995.

[4] Tim Bollerslev. Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3):307–327, 1986.

[5] Álvaro Cartea, Sebastian Jaimungal, and José Penalva. *Algorithmic and High-Frequency Trading*. Cambridge University Press, Cambridge, 2015.

[6] Rama Cont, Arseniy Kukanov, and Sasha Stoikov. The price impact of order book events. *Journal of Financial Econometrics*, 12(1):47–88, 2014.

[7] Robert F. Engle and Andrew J. Patton. Impacts of trades in an error-correction model of quote prices. *Journal of Financial Markets*, 7(1):1–25, 2004.

[8] Xavier Gabaix, Parameswaran Gopikrishnan, Vasiliki Plerou, and H. Eugene Stanley. A theory of power-law distributions in financial market fluctuations. *Nature*, 423 (6937):267–270, 2003.

[9] Olivier Guéant. Optimal market making. *Applied Mathematical Finance*, 24(2): 112–154, 2017.

[10] Olivier Guéant, Charles-Albert Lehalle, and Joaquin Fernandez-Tapia. Dealing with the inventory risk: A solution to the market making problem. *Mathematics and Financial Economics*, 7(4):477–507, 2013.

[11] Thomas Ho and Hans R. Stoll. Optimal dealer pricing under transactions and return uncertainty. *Journal of Financial Economics*, 9(1):47–73, 1981.

[12] J.P. Morgan/Reuters. RiskMetrics — Technical Document. Technical report, J.P. Morgan/Reuters, New York, 1996.

[13] Richard Roll. A simple implicit measure of the effective bid-ask spread in an efficient market. *The Journal of Finance*, 39(4):1127–1139, 1984.